Few-Shot Learning With Prototype Rectification for Cross-Domain Hyperspectral Image Classification

Anyong Qin, Chaoqi Yuan, Qiang Li, Xiaoliu Luo, Feng Yang, Tiecheng Song, and Chenqiang Gao

***Abstract*— Deep learning has been extensively applied to hyperspectral image (HSI) classification and has achieved significant success. However, the number of labeled samples available for HSI classification tasks is typically limited in practical applications, which makes the high-accuracy of HSI small-sample classification still a challenging research task. Therefore, metric-based prototypical networks for few-shot learning (FSL) have become increasingly popular. However, the majority of existing FSL methods typically have problems with biased prototypes and domain shifts. To address these issues, this article proposed a prototype rectification network framework for cross-domain few-shot HSI classification. Specifically, to obtain more representative prototypes, we designed a query-guided prototype rectification module, which can rectify the feature distribution of the support set prototype and obtain a more representative prototype for subsequent training tasks. Then, we introduced a prototype-based interclass loss function to alleviate the interclass confusion that may result from prototype rectification. Furthermore, we construct an intermediate domain between the source domain and the target domain to alleviate domain shift, which helps mitigate the difficulties of domain transfer and achieve a more comprehensive domain alignment. The experimental results on four publicly available HSI datasets demonstrate that our proposed method outperforms the existing FSL methods.**

***Index Terms*— Cross-domain, few-shot learning (FSL), hyperspectral image (HSI) classification, prototypical network.**

I. Introduction

HYPERSPECTRAL image (HSI) consists of a 3-D data cube comprising hundreds of spectral channels, which contains abundant spatial and spectral information. This information can be utilized to accurately identify the categories of objects on the ground. The purpose of HSI classification is to utilize the spectral and spatial characteristics contained in an HSI to assign each pixel to a corresponding land-cover class. Due to these characteristics, HSI classification has been widely used in many fields, such as agricultural management, environmental monitoring, and urban planning [1], [2], [3].

According to the feature extraction approach, HSI classification methods can be divided into two categories: handcrafted shallow methods and deep learning methods [1], [4]. The classification methods of the first category are generally based on manually designed features or traditional classifiers, such as principal component analysis (PCA) [5], K-nearest neighbor (KNN) [6], and support vector machines (SVMs) [7]. These methods usually obtain better classification results, but they often require manual feature extraction and are poorly adapted to different application scenarios, which is insufficient to meet the practical needs. In recent years, deep learning methods have been widely applied to HSI classification [8], [9], [10], [11]. Compared with traditional methods, deep learning algorithms can automatically learn the high-level features of the samples and optimize the parameters through backpropagation. With the advancement of research, convolutional neural networks (CNNs) have become a crucial framework for image classification tasks, and many CNN-based HSI classification methods have been proposed. Zhang et al. [12] and Li et al. [13], respectively, applied 3D-CNN to HSI classification, capturing spatial–spectral joint features of HSI data more comprehensively. However, overfitting issues can occur easily when using deeper CNN models for HSI classification. To alleviate the overfitting issue, Zhong et al. [14] proposed an end-to-end spectral–spatial residual network (SSRN) for HSI classification, which alleviates the declining-accuracy phenomenon. Zhang et al. [15] introduced a CNN-based unsupervised framework for multiscale and diverse feature learning to jointly extract deep features from HSI. Deep learning methods can enhance model performance to a certain extent, but most of their outstanding performance relies on a large number of mutually labeled samples. However, the process of collecting labeled samples is time-consuming in practice, which results in the number of labeled samples available for HSI classification tasks being usually limited. Therefore, HSI classification with a limited number of labeled samples is still a challenging research task.

Recently, few-shot learning (FSL), as a paradigm of metalearning, has garnered widespread attention for its capacity to solve the few-shot classification problem with a limited

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amount of labeled data, and the representative approaches include Siamese networks [16], matching networks [17], relational networks [18], prototypical networks [19], and so on. In recent years, metric-based FSL methods have been widely applied in HSI classification due to their simplicity and effectiveness [20], [21], [22], [23], [24], [25], [26]. Prototypical networks, as one of the common methods in the metric-based FSL method, have also garnered significant attention in the academic community. The objective of a prototypical network [19] is to learn a metric space so that the classification tasks can be accomplished by computing distances to prototype representations of each class in this space. Sun et al. [27] introduced earth mover's distance (EMD) as a metric method to explore the local structure of hyperspectral embeddings, enabling the model to focus more on local information. Xi et al. [28] combined class-covariance metrics with the Mahalanobis distance to construct a flexible decision boundary for HSI classification. Zhao et al. [29] proposed a framework that combines a multiattention transformer with active learning based on adaptive superpixel segmentation (MAT-ASSAL), which allows for the selection and enhancement of more accurate and informative features in hyperspectral data by adjusting and combining the weights of different types of attention. Zeng et al. [20] proposed a multistage network with two metrics that learn representative feature relationships from the channel and spatial dimensions. It is worth noting that due to the presence of some singularities in the dataset, which are far from the center of the category and the randomness in forming the support set (N_s samples randomly selected from the N samples of each class are used to form the support set, $N_s \ll N$), in this case, the prototypes obtained by directly averaging the support set samples are usually unrepresentative. Under the classification mechanism of prototypical networks, such biased support set prototypes can lead to misclassification of categories in HSI classification. Previous studies have ignored the existence of bias in prototypes, and the presence of such bias can affect the classification performance of the model to some extent.

Metric-based FSL methods often assume that both source and target class data come from domains with the same distribution. However, both the source and target data may originate from domains with severe domain shifts in practice [30], [31]. When there is a significant domain shift, applying a model trained on the source domain to the target domain becomes particularly challenging. In recent years, inspired by the generative adversarial network (GAN), many domain adaptation methods have been proposed to mitigate domain shift issues in the cross-domain tasks [32], [33], [34], [35]. Zhu et al. [36] introduced a deep subdomain adaptation network (DSAN) based on local maximum mean discrepancy (LMMD). DSAN learns the transfer network by aligning the subdomain distributions of specific domain-layer activation cross different domains. Cui et al. [37] proposed a method, achieved through batch nuclear-norm maximization (BNM), which enhances diversity and discriminability in predictions under few-shot conditions. Li et al. [23] employed a conditional adversarial domain adaptation strategy, in which HSI data from two different domains are mutually advanced, reducing distribution discrepancies and improving the classification accuracy in the target domain. Zhang et al. [24] utilized a graph distribution information propagation mechanism (Gia-CFSL) to perform cross-domain graph alignment at both feature and distribution levels. While the aforementioned methods have achieved significant success in addressing cross-domain issues, they have not considered the negative impact of domain-specific features in both the source and target domains [38]. Domain-specific features hinder models from learning transferable features to some extent.

To solve the abovementioned problems, this article proposes a network framework that integrates the cross-domain and FSL methods for HSI classification. First, we use the prototype rectification module to reduce distributional bias between the support set and the query set to obtain more representative prototype representations for the subsequent FSL. Second, to alleviate overlapping between the rectified prototypes of different classes, we introduce an interclass loss function based on rectified prototypes to alleviate the issue of interclass confusion. In addition, we construct intermediate domains on the classifier and the domain discriminator, respectively. Gradually reducing the distance with the intermediate domains helps alleviate the transfer difficulty, enabling the model to better learn transferable features.

The main contributions of our proposed method are given as follows.

1) We design a prototype rectification module, which drives the support set distribution closer to the query set distribution by fusing feature bias terms, providing a more accurate prototype for the subsequent model training.
2) In addition, in order to prevent the interclass confusion problem caused by the rectified prototypes, we introduce a regularization term to enhance the interclass variability of the rectified prototypes and the generalization ability of the model.
3) We propose a novel framework for constructing an intermediate domain on classifier and discriminator, which aims to achieve a more comprehensive domain alignment and more robust adversarial training for domain adaptation.

The remaining sections of this article are organized as follows. Section II presents the related works. In Section III, we provide a detailed explanation of the proposed method in this article. Section IV introduces the hyperspectral imaging (HSI) datasets that are used for both the source and target domains, and describes and conducts an analysis of the experimental results. Finally, Section V concludes this article.

II. Related Work

A. Few-Shot Learning

Traditional supervised learning methods [39], [40] usually demand a substantial number of labeled samples for training the model, which enables it to accurately classify the new data. However, acquiring a large quantity of labeled data is challenging and costly in practical applications. Metalearning has gained significant attention in the academic community in recent years because of its ability to enable models to

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generalize with few labeled samples available. Metalearning primarily consists of two stages: metatraining and metatesting, which are the core components of the metalearning framework. These stages allow models to acquire general knowledge and strategies from a diverse set of metatasks and then apply this knowledge and strategies when encountering new tasks. This training method enables metalearning to better cope with situations where the number of labeled samples is limited. FSL, as one of the methods within the metalearning paradigm, has wide applications in various domains, including computer vision, natural language processing, and speech recognition.

As an effective method for mitigating the scarcity of limited labeled data, FSL has also received significant attention in the field of HSI classification in recent years. Xue et al. [21] proposed a lightweight spectral–spatial network, which allows the model to focus on sample pairs that are difficult to identify due to contrastive loss. Liu et al. [22] proposed a deep FSL method to address the challenge of limited samples in HSI classification. This approach utilized a deep residual 3-D-CNN to extract spectral–spatial features and learned a metric space where samples from the same class are brought closer, while those from different classes are pushed farther apart. Xue et al. [11] combined a lightweight spectral–spatial Siamese network with differentiated-scale graph convolutional networks, which enables the model to capture more and better discriminative features in FSL scenarios through an adaptive weighted feature fusion strategy. Zuo et al. [41] introduced an HSI classification method based on FSL and edge-labeling graph neural network (EGNN). It overcomes challenges such as high-dimensional HSI, limited labeled samples, and high nonlinearity, resulting in improved accuracy in land cover classification.

B. Prototypical Network

In metric-based FSL methods, the prototypical network as one of the popular approaches has gained widespread research and exploration due to its simplicity and effectiveness. The prototypical network was originally proposed by Snell et al. [19]. It has achieved success in the field of computer vision due to its strong capabilities of generalization and adaptation to new tasks [42], [43].

In the prototypical network, each category is represented by a prototype vector, which is computed as the average of feature vectors from all support set examples belonging to the corresponding class. Typically, in the prototypical network, distance metrics, such as Euclidean distance or cosine distance, are commonly used to measure the similarity between the query samples and the support set prototypes. A smaller distance indicates a higher likelihood that the sample belongs to the corresponding category. In the episode-based FSL training scenario, each episode is trained based on an N-way K-shot task, and K is often a very small number ($K = 1$). Due to the randomness of the selection strategy, some samples far from the center of the category may be selected for forming the support set. Consequently, the prototypes computed by simply averaging features are usually not representative. To address this issue, recent research has attempted to improve prototype estimation by introducing strategies such as prototype restore [44] and label propagation [45]. Liu et al. [26] proposed an FSL prototype contrastive learning network (RPCL-FSL) that learns representative prototypes for few-shot classification by applying triple constraints to the support set prototype. Wang et al. [46] enhanced the ability to extract domain-invariant representations by introducing a classwise attention metric module to allow the feature extractor to better focus on the class discriminative features, in addition to combining asymmetric adversarial training with embedded domain-specific information.

C. Domain Adaptation

Domain adaptation aims to address the issue of knowledge transfer from one domain (source domain) to another domain (target domain). Because of the differences in data distribution between these two domains, applying a model trained in the source domain to the target domain typically results in poorer performance. To address this issue, researchers have proposed various methods [47], [48], [49]. One of the primary approaches in shallow domain adaptation is to align domain distributions by using feature-based methods, which involves minimizing the distance between domains, with the most commonly used distance metrics being maximum mean discrepancy and Kullback–Leibler divergence. A series of deep domain adaptation approaches, such as domain adversarial networks, have emerged with the development of deep learning. Simultaneously, domain adaptation methods have also demonstrated their effectiveness in HSI classification. Lee et al. [50] introduced a self-supervised learning framework that utilizes cross-domain CNN, enabling the learning of representations from different HSIs with distinct spectral characteristics and no pixel-level annotations. Li and Kwon [51] proposed a biclassifier determinacy maximization (BCDM) algorithm with a novel metric to simultaneously enhance classifier determinism and prediction diversity. In [52], a dual graph cross-domain FSL framework was proposed, which combines FSL with domain alignment. It employs graph optimal transport (GOT) for cross-domain alignment at both the graph feature level and the graph distribution level. Zhang et al. [53] combined FSL, semisupervised learning, and domain adaptation strategy and utilized the available data information to solve the issues of lack of labeled samples for HSI classification. In domain adaptation, domain-specific features are as important as domain-invariant features, while they often ignore the effect of domain-specific features on domain adaptation.

III. Proposed Method

The framework of the proposed method is shown in Fig. 1. It mainly consists of four modules: a feature embedding module, a cross-class bias rectification module, a loss function of interclass, and a gradual minimization of the bridge module. First, the feature extraction module maps all training samples to a learnable metric space where we use Euclidean distance to compare the similarity between the support set and the query set. To obtain a more representative representation of the prototype, we employ the prototype rectification module to correct the feature distribution between the support set and

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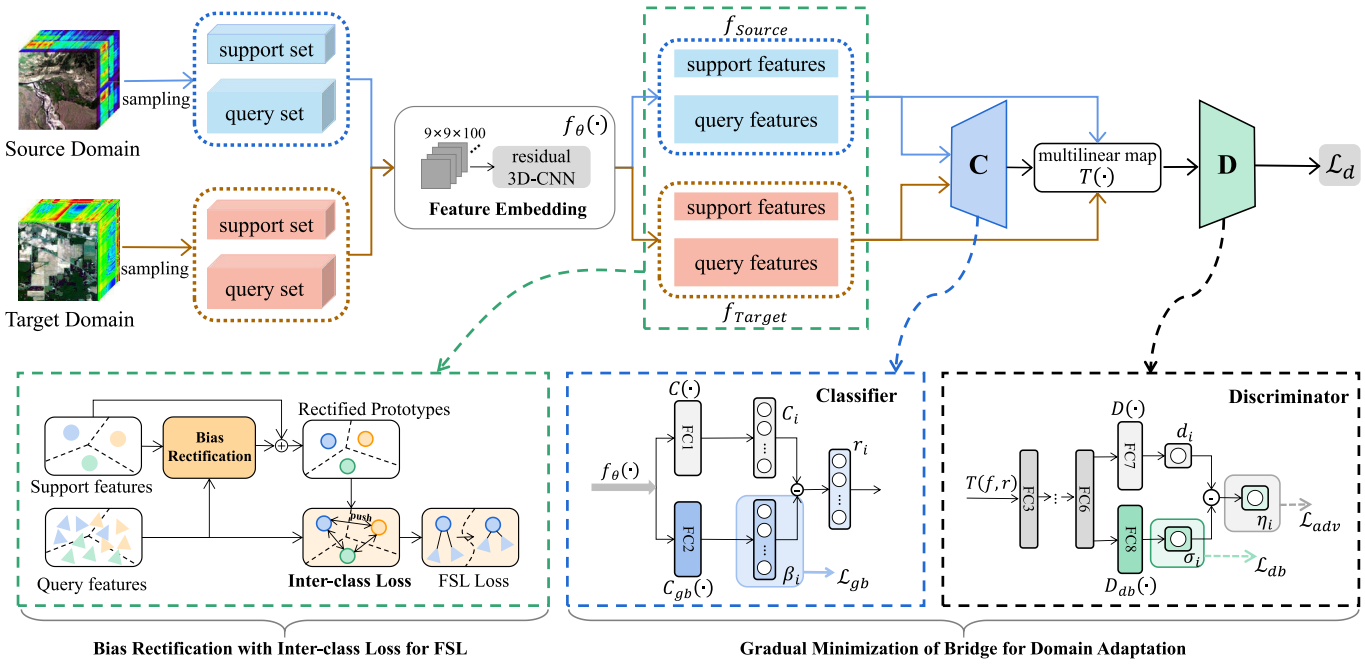


Fig. 1. Framework of the proposed method. It mainly consists of four modules: a feature embedding module, a cross-class bias rectification module, a loss function of interclass, and a gradual minimization of the bridge module. First, the construction of support and query sets is used for metatraining, and a mapping layer is used to unify the spectral bands between the source and target domains. The cross-class bias rectification module is employed to reduce the distribution discrepancy between the support set and the query set, while the interclass loss module is utilized to make the different categories rectified prototypes away from each other. The gradual minimization of the bridge module accomplishes domain alignment. (In this case, $C(\cdot)$ and $C_{gb}(\cdot)$ structures are both fully connected layers. C_i denotes the output of classifier $C(\cdot)$, $C_{gb}(\cdot)$ is used to capture domain-specific features, and its output β_i denotes the distance between the source domain (target domain) and intermediate domain. The domain-invariant feature r_i on the intermediate domain is obtained by subtracting the domain-specific feature β_i from the classification response C_i. Similarly, $D_{db}(\cdot)$ provides the additional discriminative ability for $D(\cdot)$, making the process of domain adversarial training more robust.)

the query set. In addition, to minimize the interclass confusion problem that arises after prototype rectification, we introduce a prototype-based interclass loss function. Furthermore, we construct intermediate domains in both the classifier and the discriminator. The intermediate domain constructed on the classifier is effective enough to reduce the negative impact of domain-specific features and enable the model to better learn transferable features. Similarly, the intermediate domain constructed on the discriminator ensures the robustness of adversarial training and realizes a more balanced adversarial training process.

A. Construction of Source and Target FSL

Our method is based on the episodic trained in FSL, which requires an independent selection of samples from both the source- and target-domain datasets to construct the task.

To facilitate understanding, we provide an illustrative example of how to create a task from the source domain. First, N classes are randomly selected from the source-domain data classes C_s, and then, N_s samples are randomly selected from each class as the support set $S_s = \{(x_i, y_i)\}_{i=1}^{N\times N_s}$, while N_q samples are randomly sampled from the rest of each class to form the query set $Q_s = \{(x_j, y_j)\}_{j=1}^{N\times N_q}$. Evidently, the samples in the support set are different from the samples in the query set, and each episode task consists of $N \times (N_s+N_q)$ samples.

As for the target-domain datasets, the process of forming the task in the target domain is the same as the source domain. Finally, we can obtain the target-domain support set $S_t = \{(x_i, y_i)\}_{i=1}^{N\times N_s}$ and query set $Q_t = \{(x_j, y_j)\}_{j=1}^{N\times N_q}$ for training.

B. Feature Embedding Module

The feature extraction module is responsible for extracting embedded features from the input data, which comprises a mapping layer and feature extractor. The specific structures of the mapping layer and feature extractor are elaborated on as follows.

The mapping layer is essentially a 2-D convolutional layer, consisting of d convolution kernels of size $1 \times 1 \times ch$, where ch and d denote the input and output channels of the datasets, respectively. Its purpose is to unify the dimensions of both the source- and target-domain datasets.

Then the architecture of feature extractor is shown in Fig. 2. Its primary purpose is to map the samples into a metric space, where the samples of the same class are close to each other. After obtaining the feature vectors, we will proceed to the FSL stage.

C. Cross-Class Bias Rectification

From the above part, the basic prototype P_c of class c in the support set is computed as

$$P_c = \frac{1}{N_s} \sum_{x_i \in S_c} f_\theta(x_i) \tag{1}$$

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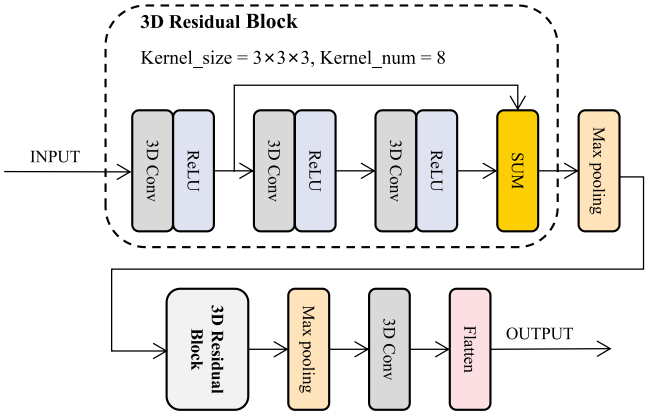


Fig. 2. Architecture of feature extractor.

where S_c denotes the cth class sample of the support set and $f_\theta(\cdot)$ represents the feature extractor with parameter θ. The prediction of the probability that the query set sample $q \in Q$ belongs to the category c is calculated as follows:

$$p_\theta(y = c|q \in Q) = \frac{\exp(-d(f_\theta(q), P_c))}{\sum_{c=1}^{N} \exp(-d(f_\theta(q), P_c))} \tag{2}$$

where Q is the query set, P_c is the prototype of the cth class, q denotes the query sample, and $d(\cdot)$ denotes a Euclidean distance function.

The prototypical network computes the prototype of each category by averaging the features of the labeled samples (support set samples). The category prediction is then made by the closest approximation of the prototype. It means that the quality of the prototype is crucial for model training. Due to the existence of samples far from the category center, in cases where the number of labeled samples is limited, the prototype obtained by directly averaging the samples of the support set may deviate from the true prototype of all the samples, which leads to a degradation of the model's classification performance. To mitigate this issue, this article introduces a bias rectification method that reduces the bias between support and query features.

The mean value, as a first-order statistical measure, can be used to represent a dataset. The support and query samples come from the same domain but are slightly different in distribution, which implies the existence of a distribution bias between them. Thus, we can measure the distribution difference between the support set and the query set by calculating the difference in their mean values. In this article, we defined a cross-class bias term δ to represent the distribution difference. The bias is formulated as

$$\delta = \mathbb{E}_{X_s \sim p_S}[X_s] - \mathbb{E}_{X_q \sim p_Q}[X_q] \tag{3}$$

where p_S and p_Q denote the distribution of the support set and the query set, respectively, while X_s and X_q represent the support set and query set features. To reduce δ at the feature level, we defined a shifting term δ_c to denote the bias between the support feature and query feature of the cth category

$$\delta_c = \frac{1}{N_s} \sum_{x_i \in S_c} f_\theta(x_i) - \frac{1}{N_q} \sum_{x_j \in Q_c} f_\theta(x_j) \tag{4}$$

where δ_c, S_c, and Q_c denote the feature shifting term and the support and query set features of the cth class, respectively.

After computing the shifting term δ, add them to the corresponding basic prototype by (5), which can make the distribution of query set and support set much closer than before

$$P_c^{\text{new}} = P_c + \delta_c. \tag{5}$$

Thus, the formula for calculating the probability of $q \in Q$ belonging to the class c can be rewritten as

$$p_\theta(y = c|q \in Q) = \frac{\exp\left(-d\left(f_\theta(q), P_c^{\text{new}}\right)\right)}{\sum_{c=1}^{N} \exp\left(-d\left(f_\theta(q), P_c^{\text{new}}\right)\right)}. \tag{6}$$

The FSL loss of source-domain data can be computed by using the true label, and the negative log probability of query samples can be computed as

$$L_{\text{fsl}}^{s} = E_{S_s, Q_s}\left[-\sum_{(x,y) \in Q_s} \log p_\theta(y = c|x)\right]. \tag{7}$$

Similarly, the FSL loss of target-domain data can be computed as

$$L_{\text{fsl}}^{t} = E_{S_t, Q_t}\left[-\sum_{(x,y) \in Q_t} \log p_\theta(y = c|x)\right] \tag{8}$$

where S_s and Q_s denote the support set and query set of the source domain, while S_t and Q_t are the support set and query set of the target domain, respectively.

D. Loss Function of Interclass

In the above section, we conducted the bias rectification on the support set prototypes, which resulted in the uncontrollable placement of the rectified support set prototypes, e.g., the prototypes of different classes would be closer together. To prevent the rectified prototypes from being close to each other and causing interclass confusion problems, we adopt the Frobenius-norm as the interclass loss to enhance the interclass separability in the embedding space, which is defined as

$$L_{\text{inter}} = \frac{2}{N(N-1)} \sum_{c=1}^{N} \sum_{\substack{j=1 \\ j>c}}^{N} \left\| \frac{P_c^{\text{new}} \cdot P_j^{\text{new}}}{\left\|P_c^{\text{new}}\right\| \left\|P_j^{\text{new}}\right\|} \right\|_F \tag{9}$$

where P_c denotes the prototype of the cth class and N denotes the number of categories.

According to (9), we can simply compute the similarity between the prototypes of each category, which can separate the prototypes of different categories from each other to some extent.

E. Gradual Minimization of Bridge for Domain Adaptation

The FSL approach allows models with limited labeled samples for training to recognize the new unseen categories. However, in cross-domain scenarios, the data distributions of the source domain D_S and target domain D_T are always different, which means that there is a domain shift between the two domains. The existence of the domain shift problem

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makes the model trained on D_S cannot be well applied to the D_T. The key to mitigating the issue of domain shift lies in acquiring domain-invariant data representations, which allows the trained model to be effectively applied from the D_S to the D_T. Thus, we aim to reduce the negative impact of domain-specific features and the difficulty of migration while aligning the global distributions of the source and target domains. Specifically, we use a conditional adversarial domain adaptation strategy and construct D_I on the classifier and the discriminator. A more balanced domain adversarial training process is achieved by gradually reducing the distance between D_S (D_T) and D_I.

1) Adversarial Loss: Since the classes of the two domains are usually different, traditional domain adaptation strategies may not be applicable to cross-domain FSL. Inspired by conditional domain adversarial network (CDAN) [34], we use a conditional discriminator $D(\cdot)$ on the source distribution $p_s(x)$ and target data distribution $p_t(x)$ to alleviate the domain shift. The result g predicted by the classifier $C(\cdot)$ contains the category discriminative information, so we condition the domain discriminator $D(\cdot)$ on the classifier's prediction g by using the joint variable (f, g) of f and g. In this case, the domain variance of the feature representation f and the classifier prediction g can be modeled simultaneously. The adversarial loss function L_{adv} on the discriminator $D(\cdot)$ can be defined as follows:

$$\min_{D} \max_{F,C} L_{\text{adv}} = -\mathbb{E}_{x_i^s \sim p_s(x)} \log\left[D(f_i^s, g_i^s)\right] - \mathbb{E}_{x_j^t \sim p_t(x)} \log\left[1\text{-D}(f_j^t, g_j^t)\right] \tag{10}$$

where $f = f_\theta(\cdot)$ denotes the feature representation and $g = C(\cdot)$ denotes the prediction of the classifier $C(\cdot)$.

The multilinear map $T_\otimes(f, g)$ is chosen to capture the complex structure of HSI features further, where $\otimes$ denotes the outer product of multiple random vectors. The dimension of a multilinear map is $d_f \times d_g$, which is too high dimensional to be embedded into deep learning models. To avoid the dimension explosion, $T_\otimes(f, g)$ can be replaced with randomized multilinear map $T_\odot(f, g) = (1/\sqrt{d})(R_f f) \odot (R_g g)$, where $\odot$ denotes the elementwise product, $d \ll d_f \times d_g$, while $R_f f$ and $R_g g$ are random matrices and d_f and d_g denote, respectively, the dimension of the feature representation f and the classifier prediction g, so the final strategy of multilinear map $T(f, g)$ is formulated as follows:

$$T(f, g) = \begin{cases} T_\otimes(f, g), & d_f \times d_g \leq 1024 \\ T_\odot(f, g), & \text{else.} \end{cases} \tag{11}$$

2) Bridge Loss on Classifier and Discriminator: Domain-specific features refer to the features that are unique to each domain and also numerously exist in respective domains. To some extent, the presence of these domain-specific features significantly hinders the alignment of data distributions between D_S and D_T. In order to obtain domain-invariant data representations better and mitigate the overall domain transfer difficulties, motivated by gradually vanish bridge (GVB) for adversarial domain adaptation [38], we introduced an easy-to-hard strategy by constructing intermediate domains on both the classifier and the discriminator, aiming to achieve better domain alignment through the domain adversarial training. Specifically, we construct intermediate domains D_I on both the classifier and the discriminator and define the notion of a "bridge," which can be used to measure the distance between the ideal representation and the true representation.

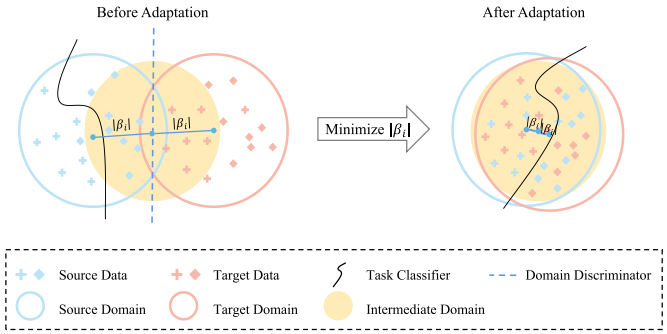


Fig. 3. Brief illustration of our domain adaptation method on the classifier. We achieve feature distribution alignment between source and target domains by constructing intermediate domains. We use $|\beta_i|$ to denote the distance between the existing representation and the ideal representation (intermediate domain). As the range of $|\beta_i|$ continues to reduce, the source and target domains will be pushed closer to the intermediate domain, which reduces the difficulty of domain adaptation and, thus, allows the model to learn more domain-invariant features.

As shown in Fig. 3, we denote the D_S, D_T, and D_I with different colored circles. We assume that D_I should lie in the middle of D_S and D_T, and there are much fewer domain-specific features in D_I than in D_S and D_T. The distance between D_S (D_T) and D_I is denoted by $|\beta_i|$, which approximates a measure of the quantity of domain-specific features in D_S (D_T). Before domain adaptation, the value of $|\beta_i|$ can be large due to the large number of domain-specific features that tend to be present in the D_S (D_T). It can lead to misclassification of categories. To minimize the effect of the domain-specific feature β_i, $|\beta_i|$ should be constantly reduced to push the D_S (D_T) closer to D_I. In this case, domain-specific features in D_I are reduced, which results in more transferable features and more comprehensive domain alignment.

Taking construction of the intermediate domain on the classifier as an example first, $C_i = C(x)$ is the output of classifier $C(\cdot)$. $C_{gb}(\cdot)$ is also a classifier, which plays the role of a bridge for capturing domain-specific features of D_S (D_T). By subtracting the domain-specific features β_i from the classifier response C_i, the transferable domain-invariant features $r_i = C_i - \beta_i$ can be obtained, which is computed by the following equation:

$$C_*(x) = C(x) - C_{gb}(x). \tag{12}$$

In order to minimize the impact of the domain-specific feature β_i, $|\beta_i|$ should be continuously reduced so that D_S (D_T) is closer to D_I. Thus, the constraint loss function of C_{gb} is computed as

$$L_{gb} = \frac{1}{(N_S + N_T) \times N} \sum_{i=1}^{N_S+N_T} \sum_{j=1}^{N} |\beta_{i,j}| \tag{13}$$

where N is the number of categories in the dataset and β_i is an N-dimensional vector. Specifically, N is set to 16 for the

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IP and SA datasets and 9 for the PC and UP datasets. N_S and N_T denote the number of samples in D_S and D_T, respectively.

Multiple discriminators can provide domain discrimination from different aspects [54]. Accordingly, the discriminator $D(\cdot)$ should be appropriately adjusted. We construct the intermediate domain D_I on the discriminator to achieve more balanced adversarial learning, $D_{db}(\cdot)$ models the measurement between the ideal decision boundary (D_I) and the discriminator $D(\cdot)$, which provides additional discriminative power for $D(\cdot)$. σ_i denotes the output of $D_{db}(\cdot)$, which can be computed as

$$\sigma_i = D_{db}(T(f, C_*(x))). \tag{14}$$

Similar to $C_{gb}(\cdot)$, we aim to minimize the range of $|\sigma_i|$ on the discriminator progressively, so the loss function of $D_{db}(\cdot)$ is defined as

$$L_{db} = \frac{1}{(N_S + N_T)} \sum_{i=1}^{N_S+N_T} |\sigma_i|. \tag{15}$$

From the above part, it is clear that the overall discriminator becomes

$$D_*(T(f, C_*(x))) = D(T(f, C_*(x))) - D_{db}(T(f, C_*(x))). \tag{16}$$

The final output η_i of the overall discriminator is computed as follows:

$$\eta_i = d_i - \sigma_i \tag{17}$$

where $d_i = D(T(f, C_*(x)))$ is the output of the discriminator $D(\cdot)$.

Then, the domain adversarial loss can be rewritten as

$$\begin{aligned} L_{\text{adv}} = &-\mathbb{E}_{x_i^s \sim p_s(x)} \log\left[D_*\left(T\left(f_i^s, C_*\left(x_i^s\right)\right)\right)\right] \\ &- \mathbb{E}_{x_j^t \sim p_t(x)} \log\left[1\text{-}\mathrm{D}_*\left(T\left(f_j^t, C_*\left(x_j^t\right)\right)\right)\right]. \end{aligned} \tag{18}$$

Therefore, the domain adaptation loss L_d can be computed as

$$L_d = L_{gb} + L_{db} + L_{\text{adv}}. \tag{19}$$

The total source-domain loss function L^s_{total} is given by

$$L^s_{\text{total}} = L_{\text{inter}} + L^s_{\text{fsl}} + L_d. \tag{20}$$

Similarly, the total target-domain loss function L^t_{total} is computed as

$$L^t_{\text{total}} = L_{\text{inter}} + L^t_{\text{fsl}} + L_d. \tag{21}$$

Finally, the total loss function for our method is given by

$$L_{\text{total}} = L^s_{\text{total}} + L^t_{\text{total}}. \tag{22}$$

IV. Experiments

A. Experimental Dataset

Because our method is a type of transfer learning, the experimental datasets are divided into two groups: the source-domain dataset and the target-domain dataset. We select Chikusei,[1] Houston2013,[2] Botswana,[3] Kennedy Space Center (KSC)[3] to form the source-domain dataset, the Indian Pines (IPs)[3], Salinas (SA)[3], Pavia Center (PC)[3], and Pavia University (UP)[3] as the target-domain datasets.

TABLE I
Brief Introduction About the Four Source-Domain Datasets

	Chikusei	Houston13	Botswana	KSC
Sensor	Hyperspec-VNIR-C	ITERS-CASI 1500	EO-1	AVIRIS
Wavelength (μm)	0.363-1.018	0.364-1.046	0.4-2.5	0.4-2.5
Spatial Size	2517×2335	349×1905	1476×256	512×614
Spatial Resolution (m)	2.5	2.5	30	18
Total labeled pixels	77,592	15,029	3,248	5,211
Numbers of Bands	128	144	145	176
Numbers of Categories	19	15	14	13

1) Source-Domain Dataset: We choose Chikusei, Houston2013, Botswana, and KSC as the source-domain datasets, and the brief introduction of the four datasets is listed in Table I.

2) Target-Domain Dataset: We select IPs, Salinas (SA), PC, and UP as the target-domain datasets in the experiments. The details of the four target-domain datasets are listed in Table II.

IP Dataset: The IPs dataset was acquired in 1992 by airborne visible infrared imaging spectrometer (AVIRIS) sensors in the USA. It consists of 224 bands ranging from 0.4 to 2.5 μm, with a spatial resolution of 20 m/pixel. After removing the spectrum covering the water absorption range, 200 bands are retained. This dataset contains a total of 145 × 145 pixels, of which 10 249 pixels are labeled. These pixels are divided into 16 classes.

SA Dataset: The Salinas dataset was captured by the AVIRIS sensor over Salinas Valley in the USA with 204 bands, which ranges from 360 to 2500 nm, and its spatial resolution is 3.7 m/pixel. There are 16 categories representing different types of land cover. It contains 512 × 217 pixels, and there are 54 129 labeled pixels in it.

PC Dataset: The PC dataset was gathered by reflective optics spectrographic image system (ROSIS) over Pavia in Italy. After removing the water absorption region, 102 bands are retained, which ranges from 430 to 860 nm, and its spatial resolution is 1.3 m/pixel. The image dimension is 1096 × 715, and it contains 148 152 labeled pixels, which are divided into nine categories.

UP Dataset: The UP dataset was collected by ROSIS in Italy. The image dimension is 610 × 340 with a 1.3-m/pixel spatial resolution and includes 103 bands. It contains a total of 42 776 labeled pixels and covers nine classes.

B. Experimental Setting

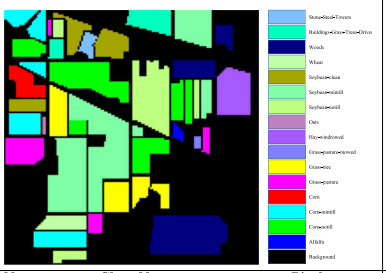
As for the data preprocessing stage of the source domain, there are 55 categories remaining after we remove the categories with labeled samples less than 200 from a total of 61 categories, which ensures that each category has a sufficient number of samples for training. Following that, the size of the spatial cube is set to 9 × 9 as the previous

[1] [Online].https://naotoyokoya.com/Download.html
[2] [Online].https://hyperspectral.ee.uh.edu/?page_id=459
[3] [Online].https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

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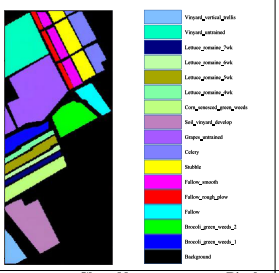
TABLE II
BRIEF INTRODUCTION ABOUT THE FOUR TARGET-DOMAIN DATASETS

Indian Pines



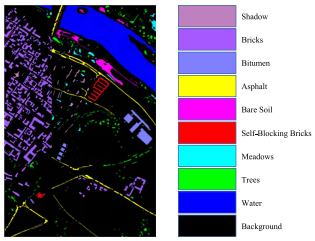
No.	Class Name	Pixels
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-tree	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
	Total	10249

Salinas



No.	Class Name	Pixels
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
	Total	54129

Pavia Center



No.	Class Name	Pixels
1	Water	65971
2	Trees	7598
3	Meadows	3090
4	Self-Blocking Bricks	2685
5	Bare Soil	6584
6	Asphalt	9248
7	Bitumen	7287
8	Bricks	42826
9	Shadow	2863
	Total	148152

Pavia University



No.	Class Name	Pixels
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Bricks	3682
9	Shadow	947
	Total	42776

few-shot HSI classification works [22], [55]. We adopt the metalearning strategy based on the episodic-training during the training phase. We use the Adam optimizer, and the learning rate is 0.001, which facilitates the network to achieve stable convergence; the number of episodes is set to 10 000, and the settings of these parameters are the same as the other methods [23], [28]. To form a few-shot classification task, we use N to denote the number of classes in an episode and N_s and N_q to denote the quantity of support set and query set of each class, respectively. In our method, N_s is set to 1, while N_q is set to 19. The number of categories N of each dataset is set to 16 for the IP and SA datasets and 9 for the PC and UP datasets.

In order to eliminate the impact of random sampling, the final experiment results are based on the average value of the results obtained from ten runs. The classification performance of all methods is evaluated with overall accuracy (OA), average accuracy (AA), and kappa coefficients (Kappa).

C. Compared With Other Methods

In order to prove the performance of the proposed method, we compare our method with three supervised methods, including SVM [7], 3-D-CNN [13], and SSRN [14], and seven FSL methods, such as S3Net [21], DM-MRN [20], DFSL + NN [22], DCFSL [23], Gia-CFSL [24], CMFSL [28], and RPCL-FSL [26]. Support vector machines (SVMs) utilize kernel methods to map nonlinear data into a high-dimensional feature space; however, they only extract the spectral information from HSI while ignoring the spatial information. 3-D-CNN utilized the 3-D convolutional neural network to extract both spectral and spatial features simultaneously. SSRN integrated the spectral and spatial information to extract classification features, which benefits the network in obtaining discriminative features for classification.

To further ensure the fairness of experiments, for the supervised methods, we select randomly five labeled samples per class for the training phase from the target-domain datasets, and the rest of the target-domain datasets samples are used for testing.

Because of these FSL methods can learn transferable knowledge from the source-domain datasets, and then utilize it to classify the unlabeled samples in the target domain, even if the distributions of the two domains are different from each other. For DFSL + NN, DCFSL, Gia-CFSL, CMFSL, RPCL-FSL, and our method, we choose Chikusei, Houston2013, Botswana, and KSC to form the source-domain dataset, and the remaining settings are the same as described in their papers. For S3Net and DM-MRN, we use a uniform patch size, i.e., 9×9, and the rest of the parameters are the same as the settings in these papers. Tables III and VI show our method performance with randomly selected five labeled samples of each class in target domain on IP, SA, PC, and UP datasets, respectively. The corresponding classification maps are shown in Figs. 4–7. The highest accuracy in each row will be highlighted in bold, and the second highest accuracy is underlined. Each experiment will be randomly run ten times, and the average results will be reported.

Table III demonstrates the results of different methods on the IP dataset. The IP dataset possesses a characteristic of class imbalance, where the number of samples in different categories varies significantly. For example, in the IP dataset, the "oats" class consists of 20 samples, while the "Soybean-mintill" class contains 2455 samples. It brings a significant challenge to the classification, even more so in the few-shot scenario. The classification results on the IP dataset show that the accuracy of our method is slightly lower than RPCL-FSL, which is due to the fact that the triple constraints imposed by RPCL-FSL on the prototype allow the model to focus

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TABLE III

CLASSIFICATION RESULTS (%) ON THE IP DATASET (FIVE LABELED SAMPLES PER CLASS)

No.	SVM [7]	3-D-CNN [13]	SSRN [14]	S3Net [21]	DM-MRN [20]	DFSL+NN [22]	DCFSL [23]	Gia-CFSL [24]	CMFSL [28]	RPCL-FSL [26]	Ours
1	82.44±13.22	92.14±4.84	92.68±1.68	**100.00±0.00**	98.29±2.90	87.17±7.98	94.91±3.78	93.90±5.69	98.78±2.25	99.51±0.98	97.56±5.00
2	34.52±7.98	40.66±9.32	**62.17±12.11**	53.15±13.56	58.34±9.30	36.52±4.49	43.17±9.49	44.24±9.88	43.05±6.90	58.86±13.63	58.03±11.81
3	37.24±12.48	49.47±6.10	52.73±10.20	61.89±6.19	53.55±11.47	52.48±4.95	52.19±8.29	55.84±6.90	56.27±6.69	65.82±10.31	**66.47±8.33**
4	58.88±14.03	61.15±15.89	79.86±9.48	**93.18±10.54**	81.94±14.25	64.56±9.30	79.67±8.16	84.09±10.26	89.31±8.27	90.73±8.13	89.22±11.47
5	69.16±6.28	74.37±8.30	80.25±9.03	79.68±0.13	77.93±8.57	70.52±8.80	75.02±8.75	72.95±7.00	76.74±8.80	80.96±8.62	**81.63±9.28**
6	65.57±12.33	80.69±9.67	91.16±4.29	85.24±3.57	**95.19±7.13**	82.47±6.12	83.64±6.60	83.72±5.01	89.78±3.38	90.79±5.47	92.73±4.36
7	87.83±6.96	97.77±4.16	99.12±1.73	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	97.83±4.01	99.57±1.30	99.13±2.61	**100.00±0.00**
8	67.57±15.24	67.42±14.09	95.06±16.17	**99.94±0.19**	94.95±6.18	75.67±12.92	81.04±14.42	84.31±12.08	90.91±12.90	92.11±9.74	85.26±14.72
9	76.00±13.06	96.67±6.83	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.50±1.50	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.33±2.00	**100.00±0.00**
10	41.03±8.01	52.05±7.37	62.83±15.15	**70.37±4.41**	65.64±14.24	52.55±8.11	61.64±10.52	65.77±9.64	61.20±9.43	69.71±6.53	64.47±8.49
11	38.07±7.47	41.35±11.05	29.99±13.97	42.41±6.43	45.83±11.35	53.17±7.93	61.54±10.47	63.94±10.26	57.69±14.43	**69.66±7.15**	67.65±5.38
12	31.33±4.20	52.55±6.14	48.25±13.15	**65.75±9.36**	60.85±12.93	31.75±4.01	50.30±11.14	43.83±5.29	55.24±10.07	63.64±12.55	59.97±11.90
13	88.00±8.59	97.93±2.23	99.66±0.47	96.25±0.93	99.25±1.10	96.15±3.89	99.00±2.01	97.70±4.38	99.65±0.59	98.10±2.34	**99.90±0.30**
14	58.60±6.96	70.65±10.85	79.65±9.28	**97.25±2.67**	87.61±7.88	80.10±6.59	84.75±5.06	86.60±9.82	87.57±8.60	91.79±6.85	91.93±7.28
15	30.60±2.03	63.51±14.71	61.96±12.69	74.01±10.69	74.70±8.70	64.66±7.01	69.16±9.36	70.16±9.99	**83.36±10.44**	82.91±11.36	78.24±7.91
16	89.77±4.13	93.85±3.84	93.50±3.42	96.05±5.33	99.43±1.16	91.72±3.97	**100.00±0.00**	97.27±5.55	99.77±0.45	99.32±1.04	99.66±0.52
OA	46.71±2.80	56.04±5.00	61.17±1.41	70.94±3.81	67.28±3.20	59.47±1.86	66.58±1.85	67.02±2.35	67.34±2.92	**75.34±2.29**	73.78±1.80
AA	59.79±1.15	70.76±3.89	76.86±2.51	82.20±1.47	80.84±1.68	71.41±1.23	77.25±0.97	77.64±1.13	80.56±1.34	**84.52±1.54**	83.30±1.95
Kappa	40.90±2.76	50.77±5.34	56.84±1.68	67.73±4.07	63.48±3.38	54.60±1.92	62.87±1.83	62.76±2.45	63.28±2.98	**72.20±2.53**	70.34±2.01

TABLE IV

CLASSIFICATION RESULTS (%) ON THE SA DATASET (FIVE LABELED SAMPLES PER CLASS)

No.	SVM [7]	3-D-CNN [13]	SSRN [14]	S3Net [21]	DM-MRN [20]	DFSL+NN [22]	DCFSL [23]	Gia-CFSL [24]	CMFSL [28]	RPCL-FSL [26]	Ours
1	96.25±1.72	95.29±2.82	96.23±2.99	99.32±1.58	98.80±1.09	97.94±1.10	98.73±0.49	98.77±1.31	98.97±1.93	98.85±1.26	**99.96±0.07**
2	92.60±6.85	97.60±1.60	99.15±1.46	99.37±1.80	99.05±2.47	97.31±1.55	99.63±1.02	99.50±0.42	**99.95±0.07**	99.88±0.15	99.85±0.20
3	83.70±5.39	96.64±3.99	95.56±5.59	93.12±5.65	**98.92±3.21**	90.63±5.77	90.73±7.40	91.71±10.35	98.83±1.45	93.58±5.61	95.60±8.70
4	99.27±0.34	92.78±4.19	95.99±5.03	99.64±0.56	**99.90±0.21**	95.78±1.41	99.45±0.75	99.10±1.01	99.50±0.50	99.15±1.66	99.23±1.02
5	93.39±3.51	93.89±2.14	91.05±4.15	95.73±3.34	**97.81±1.54**	84.07±6.29	92.98±2.99	90.39±5.36	94.77±3.51	91.60±4.36	94.09±3.46
6	97.56±1.76	90.26±6.03	99.08±0.68	99.49±0.82	**99.79±0.34**	97.04±0.80	98.85±0.15	99.25±1.02	98.20±1.62	98.75±1.20	99.60±0.76
7	99.42±0.21	94.87±8.62	99.78±0.28	99.82±0.32	**99.93±0.07**	98.11±0.67	99.68±0.71	99.26±0.92	99.39±0.43	99.56±0.45	99.84±0.19
8	58.06±8.85	70.31±12.14	77.35±19.03	80.29±6.67	70.03±9.30	67.16±8.45	75.43±6.42	80.28±5.27	76.14±10.48	78.79±7.39	**82.49±4.36**
9	94.89±2.91	89.90±13.67	97.44±5.15	99.89±0.14	**99.93±0.10**	91.43±2.11	99.58±1.03	99.08±0.97	98.13±2.50	99.79±0.40	99.83±0.40
10	78.58±10.63	74.19±10.99	84.82±17.65	85.03±4.86	**91.37±9.47**	76.22±5.90	84.00±7.30	85.05±6.42	84.08±7.28	85.29±7.14	87.12±9.28
11	88.49±5.52	95.73±2.05	96.67±4.09	99.44±0.52	**99.58±0.65**	97.89±2.64	98.16±2.98	96.48±4.13	98.80±1.55	98.82±1.51	98.86±1.94
12	97.45±2.85	88.24±8.81	85.83±13.14	97.27±2.04	98.76±0.94	98.44±1.12	**99.94±0.06**	99.44±0.89	99.56±0.58	99.54±0.76	99.69±0.34
13	97.32±0.55	87.06±5.88	92.37±6.07	96.97±3.64	98.98±0.65	97.51±5.00	99.40±0.60	98.78±1.28	98.43±1.63	99.10±0.74	**99.55±0.57**
14	91.64±2.26	82.19±10.32	93.52±5.73	89.80±12.60	92.69±9.01	96.79±3.01	99.10±0.91	96.69±3.52	**99.28±0.44**	97.44±3.66	98.70±1.90
15	54.66±3.06	58.09±14.52	61.59±20.75	**87.25±6.40**	81.36±10.48	80.88±7.17	77.54±6.86	75.25±4.99	77.78±10.22	82.45±7.80	79.72±5.36
16	80.45±2.15	93.91±5.20	93.54±3.74	**98.05±1.26**	97.08±2.65	82.65±10.87	92.39±5.48	91.23±6.67	94.34±4.73	93.12±4.80	96.17±3.58
OA	80.28±1.84	82.18±4.82	86.87±2.58	91.64±1.32	90.13±1.86	85.55±1.19	90.01±0.97	90.10±0.69	90.06±1.75	91.13±0.75	**92.03±0.83**
AA	88.22±1.44	87.56±3.07	91.25±1.64	95.03±1.31	95.25±1.33	90.62±0.96	94.10±0.83	93.77±0.94	94.76±0.80	94.73±0.98	**95.61±0.77**
Kappa	77.99±2.04	80.20±5.31	85.40±2.86	91.04±1.46	89.05±2.06	83.98±1.29	89.01±1.07	88.99±0.76	88.96±1.93	90.15±0.82	**91.13±0.92**

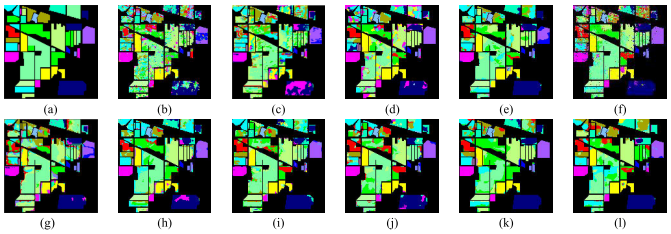


Fig. 4. Classification maps for different methods on the IP dataset. (a) Ground truth. (b) SVM. (c) 3-D-CNN. (d) SSRN. (e) S3Net. (f) DM-MRN. (g) DFSL + NN. (h) DCFSL. (i) Gia-CFSL. (j) CMFSL. (k) RPCL-FSL. (l) Proposed.

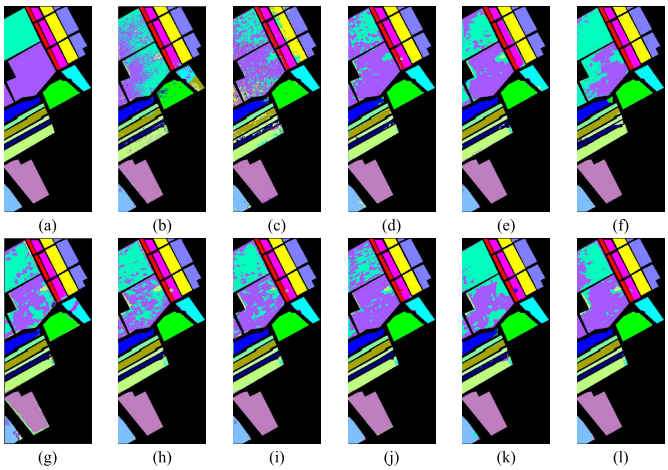


Fig. 5. Classification maps for different methods on the SA dataset. (a) Ground truth. (b) SVM. (c) 3-D-CNN. (d) SSRN. (e) S3Net. (f) DM-MRN. (g) DFSL + NN. (h) DCFSL. (i) Gia-CFSL. (j) CMFSL. (k) RPCL-FSL. (l) Proposed.

more on confusing categories, such as the second category "Corn-notill" and the fourth category "Corn." Compared with other methods, our method has the second-highest classification performance; the OA, AA, and Kappa reach 73.78%, 83.30%, and 70.34%. Especially, it surpasses the third-highest classification accuracy method CMFSL by 6.44%. The result of the experiment shows that our method has a robust generalization.

Table IV shows the classification performance results on the SA dataset. We can observe that our method's classification accuracy is higher than all of the methods from the table. It is worth noting that the classification accuracy of the eighth class "Grapes-untrained" is better than other methods, and it surpasses the second place by 2.20%. It implies that our method enables the model to better learn features, thereby making more accurate classification decisions. Furthermore, compared to the traditional supervised learning methods, such as SVM and 3-D-CNN, the FSL methods approach superior classification performance, underscores the significant potential, and prospects of FSL methods.

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TABLE V
CLASSIFICATION RESULTS (%) ON THE PC DATASET (FIVE LABELED SAMPLES PER CLASS)

No.	SVM [7]	3-D-CNN [13]	SSRN [14]	S3Net [21]	DM-MRN [20]	DFSL+NN [22]	DCFSL [23]	Gia-CFSL [24]	CMFSL [28]	RPCL-FSL [26]	Ours
1	99.15±0.21	98.71±1.16	99.71±0.46	99.30±0.65	99.58±0.76	98.99±0.47	99.80±0.16	99.74±0.20	99.76±0.17	99.51±0.30	**99.95±0.06**
2	84.48±5.69	81.24±6.60	91.28±4.01	92.58±5.12	87.07±6.31	88.07±2.49	91.04±2.01	93.15±2.79	93.06±2.32	**94.74±1.71**	92.59±3.70
3	82.44±11.98	83.07±4.52	86.30±7.46	86.76±6.92	86.60±9.13	85.41±8.55	93.11±2.31	94.06±2.48	93.87±4.27	93.89±2.82	**94.63±2.85**
4	65.36±9.74	84.94±6.66	85.20±17.60	92.97±7.22	92.49±9.14	92.61±8.76	91.73±9.71	92.37±9.87	**94.86±4.13**	91.87±6.62	93.10±8.32
5	88.59±6.47	76.49±4.97	83.70±8.96	90.42±6.41	81.77±9.87	82.47±6.91	90.65±5.44	90.16±6.24	89.42±6.55	**92.14±3.41**	89.37±6.98
6	94.22±3.17	80.88±10.20	90.45±11.15	98.98±0.61	**99.02±0.79**	88.36±1.98	96.45±2.33	94.95±4.11	94.56±3.63	97.00±1.56	97.80±1.66
7	74.03±6.43	89.44±4.49	89.40±4.72	**90.00±5.85**	86.73±3.46	77.41±9.73	85.12±3.61	86.49±3.18	87.19±6.05	86.04±3.49	88.38±3.70
8	90.33±7.88	91.69±4.56	94.68±9.10	98.24±0.83	99.21±0.68	99.16±0.54	99.14±0.46	98.85±0.56	98.52±0.43	**99.29±0.47**	99.00±0.49
9	**99.93±0.34**	92.89±6.68	98.07±1.68	96.52±2.51	98.50±1.39	87.83±5.26	99.00±1.06	99.38±0.66	99.42±0.34	96.57±1.94	99.51±0.52
OA	92.89±2.60	92.55±1.63	95.46±3.31	97.34±0.57	96.96±0.48	95.43±0.53	97.37±0.35	97.51±0.42	97.54±0.29	97.74±0.18	**97.88±0.43**
AA	86.13±4.86	86.59±1.57	90.98±3.07	93.97±1.09	92.33±1.51	89.20±1.79	94.00±0.68	94.35±1.06	94.17±0.97	94.56±0.65	**94.92±0.58**
Kappa	90.08±3.52	89.56±2.23	93.65±4.52	96.26±0.80	95.70±0.68	93.55±0.75	96.28±0.49	96.49±0.59	96.52±0.40	96.81±0.26	**97.00±0.61**

TABLE VI
CLASSIFICATION RESULTS (%) ON THE UP DATASET (FIVE LABELED SAMPLES PER CLASS)

No.	SVM [7]	3-D-CNN [13]	SSRN [14]	S3Net [21]	DM-MRN [20]	DFSL+NN [22]	DCFSL [23]	Gia-CFSL [24]	CMFSL [28]	RPCL-FSL [26]	Ours
1	62.56±4.66	62.05±10.96	66.68±22.76	80.68±7.41	79.56±10.07	83.36±8.23	79.12±7.65	76.44±11.63	75.44±13.72	80.43±9.24	**88.65±5.03**
2	60.16±15.36	61.07±9.41	82.34±6.48	73.35±9.19	73.74±13.07	**86.56±8.15**	85.48±6.72	86.33±6.55	82.92±8.56	77.74±8.93	85.26±7.44
3	55.01±18.02	72.01±10.47	75.33±10.86	70.12±9.04	**84.96±8.02**	59.15±10.01	63.13±11.77	60.96±10.44	68.14±14.08	63.04±14.13	61.89±14.18
4	81.57±6.47	78.56±5.84	82.90±11.33	87.01±5.08	90.97±5.64	92.87±6.21	92.07±4.02	92.87±4.18	**93.51±3.28**	91.95±5.00	93.25±4.82
5	94.81±7.00	98.86±1.59	99.38±1.02	99.49±1.21	99.57±0.78	**99.93±0.14**	99.72±0.54	99.07±1.54	99.81±0.36	99.29±1.24	99.79±0.39
6	53.85±17.05	70.65±9.31	74.74±7.85	73.47±12.84	81.47±12.71	53.90±12.45	71.07±10.23	73.23±10.00	**86.39±11.72**	76.83±11.50	71.79±10.05
7	87.93±4.62	88.80±4.53	93.09±8.50	**99.02±0.40**	98.66±0.74	61.21±17.47	79.17±7.63	82.22±6.96	82.44±8.10	80.06±13.19	81.06±9.84
8	67.39±9.52	53.94±7.12	51.13±17.95	77.73±9.46	75.37±13.24	64.64±7.58	69.97±16.98	71.16±17.72	67.47±10.41	**83.81±6.75**	78.29±13.24
9	**99.85±0.11**	94.29±2.85	98.06±3.17	90.28±10.45	99.80±0.27	93.37±9.63	97.60±3.12	98.28±1.88	99.33±0.90	94.30±7.15	97.25±6.07
OA	64.52±5.01	66.29±5.18	77.24±3.19	81.72±4.07	79.63±6.18	76.44±2.57	81.39±2.93	82.10±2.91	81.75±3.30	82.28±2.92	**83.62±1.84**
AA	73.68±1.39	75.58±3.26	80.41±3.06	83.46±2.17	83.12±2.40	77.22±2.39	81.93±1.94	82.29±1.97	83.94±1.34	83.05±2.26	**84.13±1.87**
Kappa	55.84±4.85	58.35±5.69	70.64±3.94	76.75±4.72	74.42±7.11	69.67±2.87	76.24±3.58	77.02±3.64	76.64±3.84	77.39±3.32	**78.72±2.07**

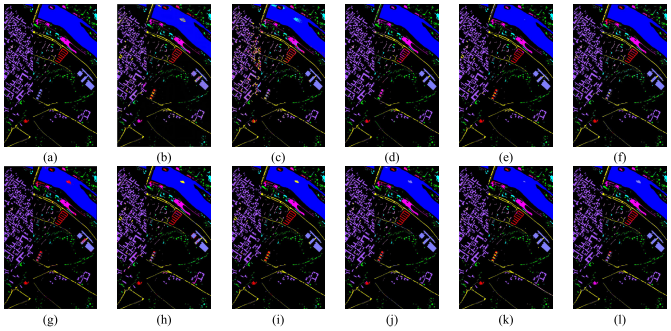


Fig. 6. Classification maps for different methods on the PC dataset. (a) Ground truth. (b) SVM. (c) 3-D-CNN. (d) SSRN. (e) S3Net. (f) DM-MRN. (g) DFSL + NN. (h) DCFSL. (i) Gia-CFSL. (j) CMFSL. (k) RPCL-FSL. (l) Proposed.

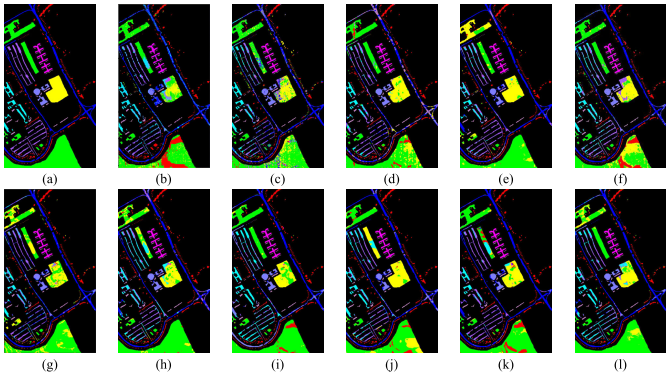


Fig. 7. Classification maps for different methods on the UP dataset. (a) Ground truth. (b) SVM. (c) 3-D-CNN. (d) SSRN. (e) S3Net. (f) DM-MRN. (g) DFSL + NN. (h) DCFSL. (i) Gia-CFSL. (j) CMFSL. (k) RPCL-FSL. (l) Proposed.

Tables V and VI provide the classification performance of different methods on UP and PC datasets, respectively. From the table, it is evident that our results remain the best among these methods. Comparing the classification results, we can conclude that the FSL method is better equipped to handle scenarios with fewer labeled samples. In comparison among these methods, the domain adaptation technique has a positive impact on the cross-domain problem, which can alleviate the domain shift to some extent. The experimental results of the classification of various methods once again demonstrate that the proposed method not only has stronger robustness but also greater generalization capabilities in cases of limited labeled samples.

To further verify the effect of different numbers of labeled samples on each FSL method, we also randomly choose one, two, three, four, and five labeled samples of each category in the target domain to form the few-shot task. Each experiment will be run randomly ten times, and the average results are reported in Fig. 8. It is easy to observe that as the number of labeled samples increases, the accuracy of classification also generally improves. The accuracy of our method is slightly lower than RPCL-FSL on the IP dataset; we consider that RPCL-FSL allows the model to obtain more representative prototypes on some easily confused categories (such as the second category "Corn-notill" and the fourth category "Corn") by imposing triple constraints. In addition, the accuracy of our method consistently ranks first among these compared methods on the SA, PC, and UP datasets.

Furthermore, we also demonstrated five cross-domain FSL methods (DCFSL, Gia-CFSL, CMFSL, RPCL-FSL, and Ours) with t-distributed stochastic neighbor embedding (t-SNE) on the IP dataset. Fig. 9 displays DCFSL, Gia-CFSL, CMFSL, RPCL-FSL, and our method separately. As can be seen in the figure, the intraclass clustering between class 11 "Soybean-mintill" and class 12 "Soybean-clean," as well as between class 5 "Grass-pasture" and class 7 "Grass-pasture-mowed" of our method, is tighter, while the interclass dispersion is larger.

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TABLE VII
CLASSIFICATION RESULTS (%) ON THE HOUSTON2013 DATASET (FIVE LABELED SAMPLES PER CLASS)

No.	SVM [7]	3D-CNN [13]	SSRN [14]	S3Net [21]	DM-MRN [20]	DCFSL [23]	Gia-CFSL [24]	RPCL-FSL [26]	Ours
1	81.54±6.49	75.70±4.98	78.91±6.73	81.05±9.81	70.19±9.87	82.83±1.31	83.53±3.97	81.99±0.84	**84.45±4.15**
2	**92.40±6.29**	73.34±11.52	79.73±9.59	79.72±13.64	81.73±2.90	89.59±10.21	89.16±8.83	87.57±11.61	90.52±6.95
3	98.63±0.46	**99.98±0.06**	99.50±0.90	99.15±0.59	96.68±1.61	95.72±1.90	92.81±4.79	94.26±3.23	94.06±3.85
4	95.28±2.60	82.24±6.54	87.19±4.46	90.26±6.55	81.99±4.53	95.32±1.93	94.32±3.87	94.48±3.98	**96.96±2.79**
5	88.97±8.95	91.65±10.59	97.72±4.44	97.40±2.07	**99.87± 0.37**	96.83±2.58	95.37±6.01	96.60±3.31	96.43±4.15
6	78.32±1.25	85.51±7.51	83.02±7.70	**86.50±7.21**	82.34±8.35	75.66±7.72	66.50±8.18	80.14±7.25	75.45±5.54
7	66.09±8.92	63.02±12.71	67.49±14.10	72.32±11.77	70.99±8.27	72.30±10.62	74.90±12.58	**83.61±8.73**	83.45±11.80
8	31.90±0.86	20.02±9.81	42.71±0.77	35.51±13.41	40.16±14.88	51.34±13.74	50.04±11.41	40.71±20.74	**53.95±18.87**
9	51.65±7.27	41.88±10.94	45.24±16.09	55.73±14.37	60.91±10.77	**65.06±5.27**	63.31±3.98	61.88±10.76	64.02±8.61
10	52.66±5.36	35.80±8.88	44.34±17.48	44.31±10.80	59.01±15.62	49.79±17.78	46.25±17.22	48.63±21.34	**61.54±17.24**
11	69.26±1.65	60.70±11.34	65.58± 20.99	76.76±15.80	**78.65±19.61**	62.64±11.76	60.16±13.22	64.20±7.92	67.60±17.67
12	29.78±2.28	41.21±8.31	38.27±9.02	44.02±11.44	**62.63±7.96**	56.50±13.50	55.10±12.51	58.07±11.88	59.75±15.82
13	20.32±6.92	47.11±14.85	66.04±11.27	73.96±8.72	79.77±10.78	77.30±10.34	76.81±8.32	**90.63±2.40**	81.40±6.41
14	94.62±5.86	96.66±6.00	99.75±0.52	**100.00±0.00**	99.72±0.60	81.78±6.80	88.99±11.12	82.96±8.67	81.86±9.52
15	97.44±2.50	97.89±1.60	99.83±0.27	**100.00±0.00**	99.98±0.06	91.69±12.36	84.84±12.26	98.96±1.68	94.19±7.44
OA	68.27±3.35	66.70±2.31	68.55±4.47	71.36±2.16	74.85±2.05	74.37±2.89	73.15±2.47	74.61±2.29	**77.64±1.47**
AA	69.93±2.62	67.51±2.17	73.02±3.58	75.78±1.49	77.64±2.04	76.29±2.24	74.81±2.15	77.65±1.91	**79.04±1.38**
Kappa	65.60±3.60	63.51±2.52	65.93±4.79	68.97±2.34	72.72±2.22	72.22±3.12	70.90±2.65	72.50±2.48	**75.74±1.59**

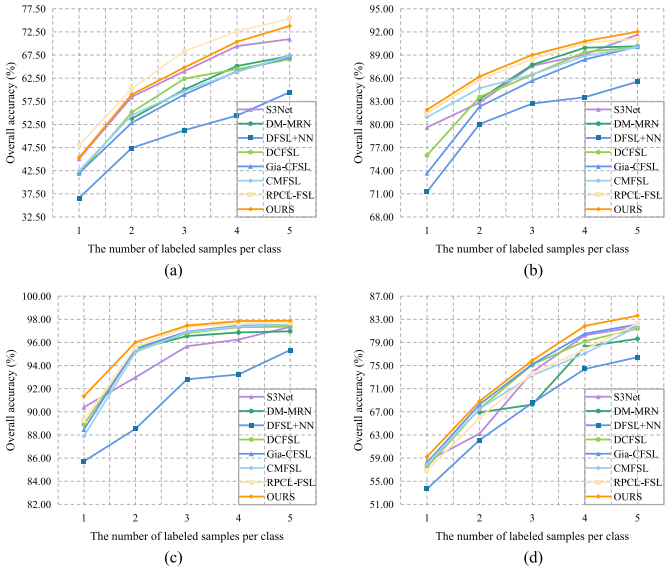


Fig. 8. Classification accuracy with a different number of labeled samples on four target-domain datasets. (a) IP. (b) SA. (c) PC. (d) UP.

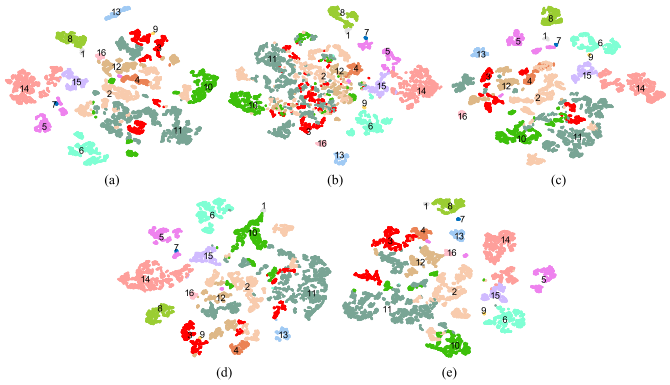


Fig. 9. Feature visualization of the IP dataset with the t-SNE. (a) 2-D feature visualization of DCFSL. (b) 2-D feature visualization of Gia-CFSL. (c) 2-D feature visualization of CMFSL. (d) 2-D feature visualization of RPCL-FSL. (e) 2-D feature visualization of the proposed method.

It is clear that, compared to the other methods, our method can effectively distinguish some confusing categories in the IP dataset.

To prove the superiority of the proposed method further, we conducted experiments on the Houston2013 dataset (HS) [56], which is a challenging HSI classification task. It was acquired by ITRES CASI-1500 in Houston in 2013 with 144 bands ranging from 364 to 1046 nm and a spatial resolution of 2.5 m/pixel. The dimensions of this image are 349 × 1905 pixels and contain 15 categories. It is worth noting that HS had fixed training and test sets corresponding to sample sizes of 2832 and 12 197 respectively.

Specifically, for SVM, 3D-CNN, SSRN, S3Net, and DM-MRN methods, we randomly selected five labeled samples per class from the 2832 samples in the training set, and 12 197 samples in the test set were used for testing. For the cross-domain FSL methods (DCFSL, Gia-CFSL, RPCL-FSL, and Ours), we choose Chikusei as the source-domain dataset and HS as the target-domain dataset. We randomly select five labeled samples per class from 2832 samples in the training set for the training phase and 12 197 samples in the test set for the testing phase. Since HS is considered the source-domain training data in the DFSL + NN and CMFSL, it is not possible to validate their classification performance on the HS dataset. In addition, all experiments were conducted ten times in order to eliminate the effect of random sampling. Table VII shows the classification accuracy of the different methods on the HS. As can be seen from Table VII, the FSL methods perform better overall than the supervised methods (such as SVM, 3D-CNN, and SSRN). This is because the FSL methods enable the model to learn more general feature representations and have a greater advantage in identifying new categories when the number of available labeled samples is limited. Compared to the other FSL methods, the proposed method ranks first in OA, AA, and Kappa, which are 2.79%, 1.39%, and 3.02% higher than the second place, which further demonstrates the superiority and robustness of the proposed method.

D. Ablation Study

1) Effectiveness of the Different Modules: To validate the effectiveness of the different modules, we conducted ablation experiments on four target domains. Specifically, we add each of these modules to the baseline network and then analyze the

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TABLE VIII
CLASSIFICATION RESULTS (%) OF THE ABLATIONS ON THE PROPOSED MODULES

	IP			SA			PC			UP		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
baseline	66.58±1.85	77.25±0.97	62.87±1.83	90.01±0.97	94.10±0.83	89.01±1.07	97.37±0.35	94.00±0.68	96.28±0.49	81.39±2.93	81.93±1.94	76.24±3.58
V1	71.13±2.34	81.58±1.18	67.43±2.53	91.29±0.81	95.05±0.93	90.31±0.90	97.78±0.28	94.65±0.63	96.87±0.39	82.06±1.72	83.27±1.84	76.70±2.00
V2	69.42±1.95	82.01±1.64	66.02±2.17	91.31±0.76	94.95±0.79	90.38±0.84	97.38±0.45	94.30±0.63	96.30±0.64	82.30±2.06	**83.77±1.10**	77.59±2.39
V3	69.07±2.11	79.75±1.21	65.05±2.15	91.32±0.82	95.25±0.66	90.39±0.90	97.49±0.34	94.12±0.88	96.45±0.48	82.13±2.32	83.47±1.31	76.94±2.45
V1+V2	72.47±3.60	82.49±3.60	68.93±3.93	91.78±0.80	95.42±0.87	90.86±0.89	97.79±0.93	94.74±0.93	96.87±0.55	**83.01±2.47**	83.64±2.33	**78.09±2.88**
V1+V3	72.35±3.32	82.65±1.89	68.81±3.62	91.79±0.44	**95.52±0.6**4	90.87±0.49	**97.81±0.47**	94.71±0.67	96.88±0.66	82.79±3.15	83.50±1.60	77.78±3.65
V2+V3	**73.32±2.64**	**82.87±2.35**	**69.82±3.03**	**91.89±0.82**	95.38±0.67	**90.98±0.92**	97.80±0.33	**94.82±0.58**	**96.94±0.47**	82.85±2.58	83.63±1.75	77.99±3.08

performance by comparing the results before and after the modules are added. We use $V1$, $V2$, and $V3$ to represent the network after adding the cross-class bias rectification module, the gradual minimization of the bridge module, and the interclass loss module to the baseline, respectively.

The classification results of the ablation experiments on the four datasets are illustrated in Table VIII. By conducting the analysis of the results, the conclusions can be drawn as follows.

The first line $V1$ shows the results of adding the bias rectification module to the baseline network on the four datasets. Compared with the baseline network, our method improves the AA by 4.33%, 0.95%, 0.65%, and 1.34% on the four datasets, respectively. The prototype obtained at present may have a large deviation from the actual prototype of this category because of the randomness of sample selection. The experimental results show that this module can greatly reduce the difference in distribution between the support set and the query set.

The second line $V2$ shows the classification results of four datasets after adding gradual minimization of the bridge module to the baseline. Especially in the case of IP and SA datasets, this module improves OA by 2.84%–1.30% compared with the baseline. It means our method can enable the model to learn domain-invariant data representations better and effectively alleviate the issue of domain shift.

The third line $V3$ demonstrates the results of adding the interclass module to the baseline on four datasets. In comparison to the baseline network, our approach shows a certain degree of improvement across the OA, AA, and Kappa. It is particularly pronounced on the IP dataset; there have been improvements of 2.49%, 2.50%, and 2.18% on OA, AA, and Kappa, respectively. The results indicate that our module can alleviate the issue of interclass confusion to a certain extent, thereby improving classification accuracy.

In addition, we explored the impact of combining different modules on the model's classification performance, as shown in rows 5–7 of Table VIII. The result shows that the three proposed modules can improve the model's generalization ability in situations with limited labeled samples.

E. Influence of Choosing Different Source-Domain Datasets on Model Performance

In order to explore the significance of the source-domain dataset selection, we chose Loukia, Botswana, Chikusei, and "C + H + B + K" as the source-domain datasets, respectively. The Loukia dataset was collected by the EO-1 Hyperion sensor. The image dimension is 249 × 945 with 30-m spatial resolution and includes 176 bands. It contains a total of 13 503 labeled pixels and covers 14 classes. Table IX demonstrates the classification results on different source-domain datasets. "C + H + B + K" refers to the dataset consisting of Chikusei, University of Houston 2013, Botswana, and KSC, where the total number of classes involved in training is 55 and the total number of samples used to construct the training task is 11 000 (each class contains 200 samples). If the source domain is Botswana or Loukia, the total number of categories available for training is 11 and the total number of training samples is 2200. When the source domain is Chikusei, there are 18 categories available for training and the total number of training samples available is 3600. When the source-domain dataset is "C + H + B + K," the classification performance is superior to using only a single dataset as the source domain. When a single dataset is used as the source domain (such as Chikusei, Botswana, or Loukia), richer categories will also improve model performance. It can be seen that the larger dataset "C + H + B + K" covers more categories, such as multiple types of soil, lands, marshes, plants, water, and man-made objects, which can provide richer information for the model to learn the transferable features better and improve the generalization ability and performance in the target domain.

TABLE IX
CLASSIFICATION RESULTS (%) OF FOUR TARGET-DOMAIN DATASETS ON THE DIFFERENT SOURCE DOMAIN

Target Dataset	Metrics	Source Dataset			
		Loukia	Botswana	Chikusei	C+H+B+K
IP	OA	69.34±3.70	70.38±2.72	71.15±1.72	**73.78±1.80**
	AA	79.33±2.67	80.32±2.38	81.07±1.61	**83.30±1.95**
	Kappa	65.39±3.97	66.50±3.01	67.33±1.89	**70.33±2.01**
SA	OA	90.58±1.19	91.08±0.95	91.52±0.55	**92.03±0.83**
	AA	94.26±0.98	94.42±1.07	95.15±0.62	**95.61±0.77**
	Kappa	89.52±1.32	90.06±1.06	90.56±0.61	**91.13±0.92**
PC	OA	96.70±0.51	97.65±0.30	97.68±0.46	**97.88±0.43**
	AA	93.40±1.21	94.47±0.92	94.49±0.75	**94.92±0.58**
	Kappa	95.43±0.55	96.70±0.43	96.72±0.65	**97.00±0.61**
UP	OA	80.96±2.95	81.06±2.80	81.84±2.54	**83.62±1.84**
	AA	82.13±2.16	82.53±2.36	82.91±1.99	**84.13±1.87**
	Kappa	75.44±3.31	75.46±3.24	76.64±2.97	**78.72±2.07**

F. Computational Complexity

We compared the running times, FLOPs, and parameters of different algorithms. The specific details are shown in Table X. Our method ranked tenth in terms of running speed among the 11 methods on all four datasets. Due to the presence of deep residual networks, both DCFSL and our method exhibit higher FLOPs and parameters. However, in situations with test times close to each other, considering the classification performance,

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TABLE X
PARAMETERS, FLOPS, AND COMPUTATIONAL TIME WITH DIFFERENT METHODS ON FOUR TARGET-DOMAIN DATASETS (FIVE LABELED SAMPLES PER CLASS)

	Method	SVM	3-D-CNN	SSRN	S3Net	DM-MRN	DFSL+NN	DCFSL	Gia-CFSL	CMFSL	RPCL-FSL	OURS
IP	Training (s)	0.70	62.35	145.28	24.42	178.15	350.03	1351.40	3641.49	964.97	443.16	1461.56
	Testing (s)	0.22	1.50	7.60	10.60	19.49	1.48	1.65	1.52	0.62	1.49	1.47
	Params (M)	-	0.10	0.35	0.09	0.18	0.03	4.26	0.31	0.77	0.07	4.26
	FLOPs (M)	-	0.40	1.73	29.61	25.86	41.51	47.35	41.78	29.57	43.18	47.36
SA	Training (s)	0.69	63.03	83.58	24.11	198.25	1066.40	1427.94	3815.48	994.19	543.96	1516.89
	Testing (s)	2.49	8.36	41.43	58.32	94.82	7.37	7.90	10.05	3.18	7.87	7.95
	Params (M)	-	0.10	0.40	0.09	0.18	0.03	4.26	0.31	0.77	0.07	4.26
	FLOPs (M)	-	0.21	1.76	29.61	25.86	41.51	47.38	41.78	29.57	43.21	47.39
PC	Training (s)	0.67	54.39	66.49	61.56	217.39	1040.30	981.47	2793.29	731.15	518.02	1055.05
	Testing (s)	6.02	21.50	106.37	151.46	154.41	30.50	18.54	20.94	8.90	18.81	19.79
	Params (M)	-	0.05	0.20	0.05	0.17	0.03	4.25	0.31	0.77	0.06	4.25
	FLOPs (M)	-	0.20	0.98	11.93	14.64	41.51	46.56	41.78	29.57	42.38	46.57
UP	Training (s)	0.68	53.86	66.56	26.01	118.71	481.66	845.46	2492.52	654.87	310.90	917.27
	Testing (s)	0.47	6.23	30.32	44.96	45.95	11.43	5.45	5.66	2.55	6.30	5.46
	Params (M)	-	0.05	0.20	0.05	0.18	0.03	4.25	0.31	0.77	0.06	4.25
	FLOPs (M)	-	0.20	0.99	11.93	14.64	41.51	46.57	41.78	29.57	42.39	46.58

the running speed and overall FLOPs of our proposed method are generally within a reasonable range and acceptable.

V. CONCLUSION

In this article, we propose a prototype rectification network framework for cross-domain few-shot HSI classification. First, to mitigate the bias between the support set prototypes and the query set samples, we design a prototype rectification module to make the support set distribution closer to the query set, effectively alleviating the issue of prototype instability. Second, to prevent the overlapping of support set class prototypes caused after prototype rectification and alleviate the interclass confusion, we introduce a prototype-based interclass loss function to alleviate these problems. In addition, we construct an intermediate domain on the classifier and the discriminator, respectively, aiming to alleviate domain shift and achieve more comprehensive and robust domain alignment. The classification results on four public datasets validate the feasibility of our proposed method.

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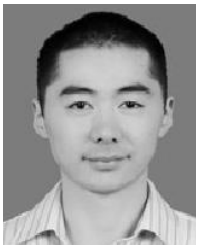
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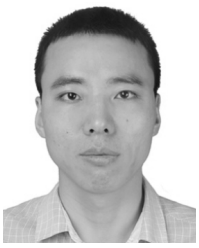
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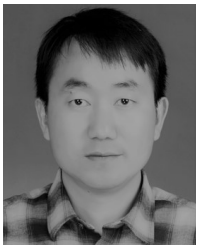
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